

AHMAD BILAL

AI & Backend Engineer

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Education

Bachelor of Science in Software Engineering — CGPA: 3.57

Sep 2023 – Present

PUCIT (Punjab University College of Information Technology), Lahore

FSc. Pre-Engineering

Nov 2021 – Jul 2023

Forman Christian College, Lahore

Professional Experience

Backend Developer — CodeBryx - Development & Technology Services

Apr 2025 – Present | Lahore

- Designed and built production-grade Python backends using FastAPI, integrating Google Maps, Places, and Hotels APIs for intelligent location-based features.
- Developed Pelopilex, a multi-step consultation and booking platform with dynamic form logic, Stripe API checkout sessions, webhook-driven payment confirmation, and payment status management.
- Built Smart Tripster, an AI-powered travel planning backend, optimizing third-party API usage through caching and efficient request batching — reducing external API costs by approximately 32%.
- Architected scalable RESTful APIs across multiple products, translating product requirements into maintainable, well-documented Python services.
- Contributed to tgs-agent-be, a FastAPI/SQLAlchemy/PostgreSQL platform, building workspace branding and pricing configuration APIs with RBAC support.
- Collaborated with cross-functional teams including frontend developers to ensure clean backend-frontend API contracts; maintained API specs and shared tooling conventions.

Backend Development Intern — CodeBryx - Development & Technology Services

Nov 2024 – Apr 2025 | Lahore

- Contributed to CloudUnity, a multi-cloud management platform spanning 3 providers (AWS, Azure, GCP) — building resource listing, tag management, and deletion workflows in Python/Flask.
- Built cost optimization recommendation pipelines for AWS and Azure: analyzed cloud spend over configurable time windows and surfaced actionable savings recommendations.
- Optimized cost optimization endpoint response times from 6 minutes to 1 minute — an approximately 83% reduction — through improved data processing and query efficiency.
- Worked with PostgreSQL and SQLAlchemy for data storage and querying; wrote clean, testable Python code and contributed to shared internal libraries.

Projects

Smart Tripster — AI-Powered Travel Planning Backend

- FastAPI backend integrating Google Maps, Places, and Hotels APIs for location-based search and intelligent travel recommendations; includes data transformation and storage pipelines.

Data Extraction & Cost Analysis Pipeline

- Built analytical pipelines to extract, process, and transform cloud usage data from AWS and Azure APIs; surfaced structured cost optimization reports with actionable savings estimates.

WebSocket-Based Real-Time Chat App

- Built with NestJS and WebSockets for real-time two-way messaging; includes a full friend-request and connection management system — demonstrates strong backend architecture and API design.

Emotion Recognition System — ML / Deep Learning

- Trained and evaluated multiple models (XGBoost, Neural Networks, RNNs, Random Forest) for facial emotion classification; documented experiments and iterated on architectures.

Certifications

Machine Learning Specialization

Sep 2024

- Stanford Online / DeepLearning.AI (Coursera) — 3-course specialization: Supervised Machine Learning (Regression & Classification), Advanced Learning Algorithms, Unsupervised Learning, Recommenders & Reinforcement Learning — taught by Andrew Ng.

Technical Skills

- Languages:** Python (primary), C, C++, C#
- Backend Frameworks:** FastAPI, Django, Flask, NestJS — REST API design, WebSockets, webhook handling
- Databases:** PostgreSQL, MySQL, SQLAlchemy, ERD design, Alembic migrations
- Testing & Workflow:** Git & GitHub, pull request review, Agile/Scrum collaboration
- Data & AI/ML:** Pandas, NumPy, XGBoost, PyTorch, scikit-learn; LLM APIs (Anthropic Claude, OpenAI)
- Frontend Awareness:** Basic React.js / Next.js for backend-frontend integration collaboration
- Tools:** Postman, VS Code, Visual Studio